Jet Propulsion Laboratory
California Institute of Technology

Université Concordia University

Polytechnique Montréal

Model-Based Systems Engineering: A Survey of Practitioner Experiences and Challenges



Maged Elaasar
NASA JPL
maged.e.elaasar@jpl.nasa.gov



Bentley Oakes
Polytechnique Montreal
bentley.oakes@polymtl.ca



Wahab Hamou-Lhadj
Concordia University / NASA JPL
wahab.hamou-lhadj@concordia.ca



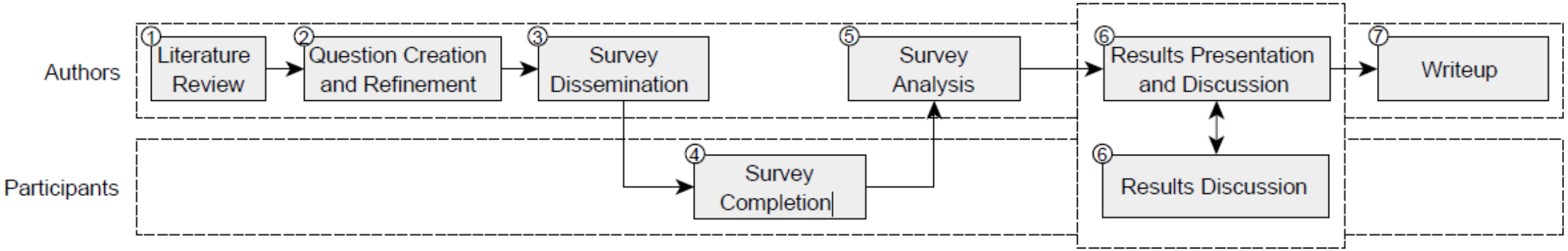
Mohammad Hamdaqa
Polytechnique Montreal
mhamdaqa@polymtl.ca

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Goal of the study

- MBSE enhances collaboration and system understanding by using models.
- Key benefits include improved traceability, error reduction, early issue detection, and efficiency.
- Despite its promise, MBSE adoption is inconsistent, with limited real-world data on standardized use across large organization.
- This study surveyed 76 participants and analyzed practices, tools, challenges, and recommendations

Survey Process



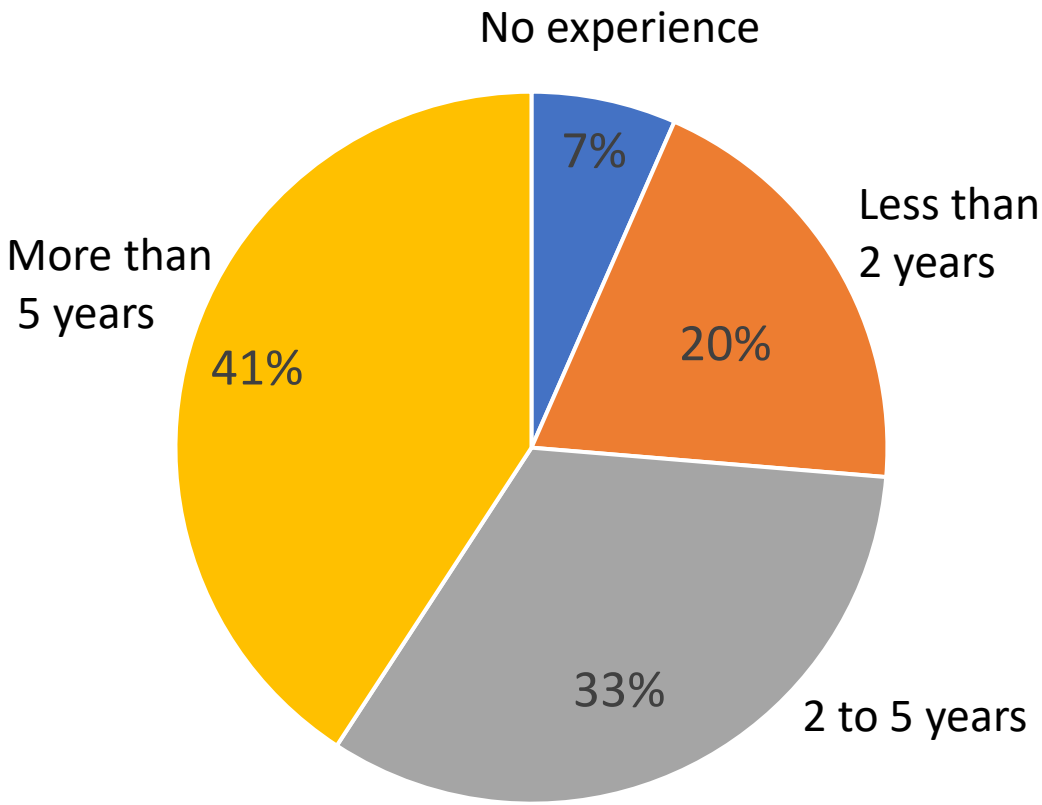
76 participants

onto:Nexus workshop 2024 involving 120+ attendees

Survey Questions

1. **Participants' Background:** Home institution, level of experience, industries, experience with MBSE tools, etc.
2. **MBSE Methods and Tools:** level of MBSE practice, MBSE method followed, scope of MBSE projects, tool customization, domain integration, collaboration.
3. **MBSE Model Management and Evolution:** stakeholders visibility, MBSE change process, model co-evolution, change impact, across-organization collaboration, other challenges.

Background of the Participants

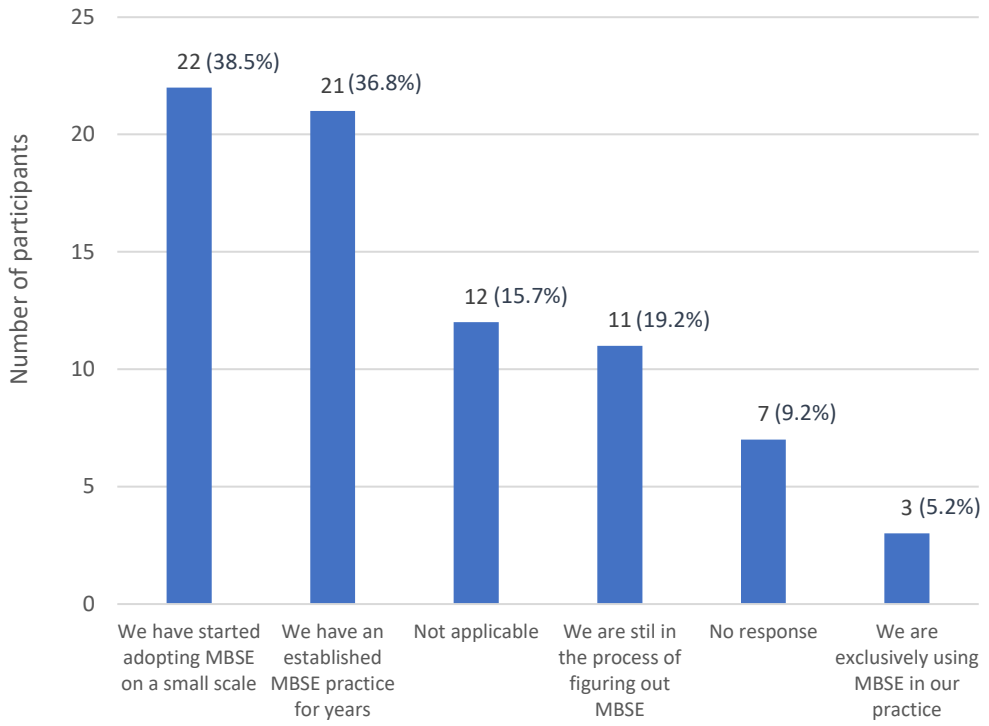


Level of experience

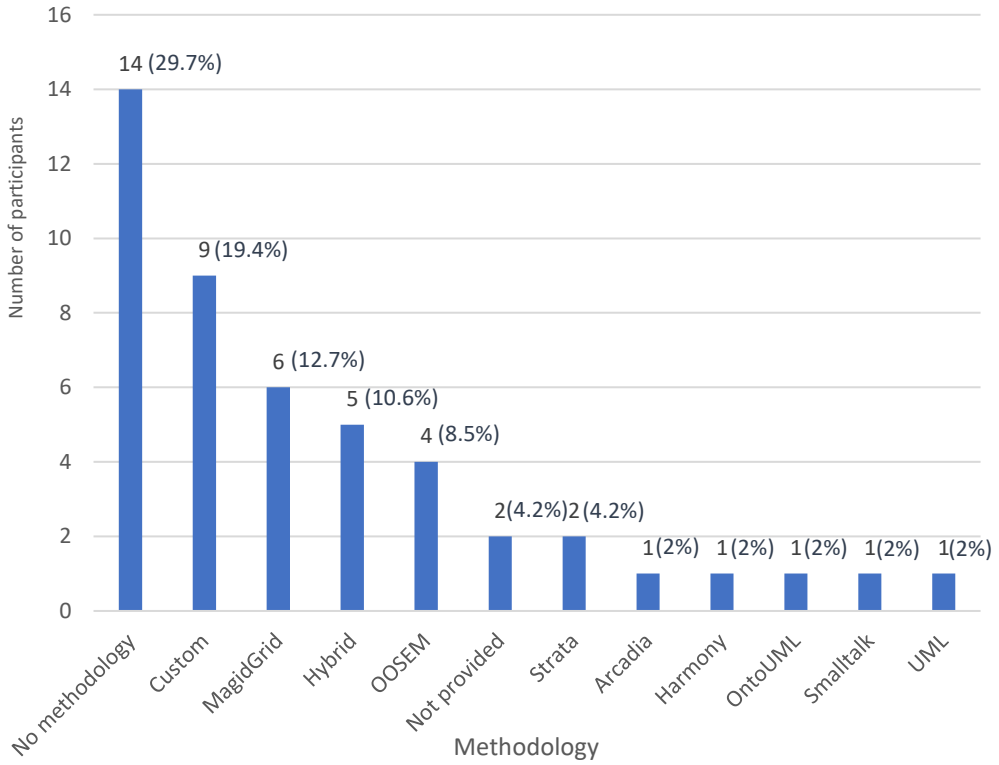
- North America (58%), Europe (23%), Asia and others (18%)
- 51% of participants identified as researchers, 45% as systems engineers, methodologists (24%), managers (21%), and tool experts (11%).
- 60% of the participants are from companies in heavy industries, aerospace (53%), defense (36%), automotive (22%), and aviation (14%).
- 38% identified as being in academia.

Participants' profile

MBSE Methodologies and Level of Practice



Level of Practice



MBSE Methodologies

Feedback from the Participants

- Examples of comments:
  - “all MBSE methodologies need to be tailored to the needs of the projects”
  - “lack of training and established best practices”

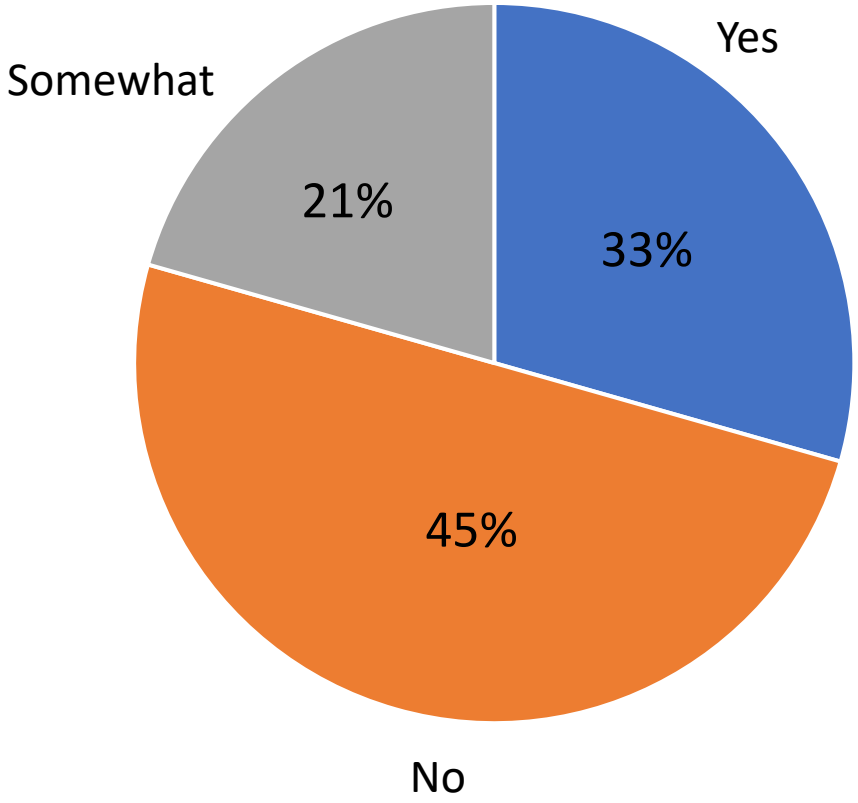
- Most of the participants stated that they tend to customize and/or combine existing methodologies to meet the needs of their projects.

- MagicGrid, OOSEM, and a hybrid approach emerged as the most popular options.

Methodology Scope

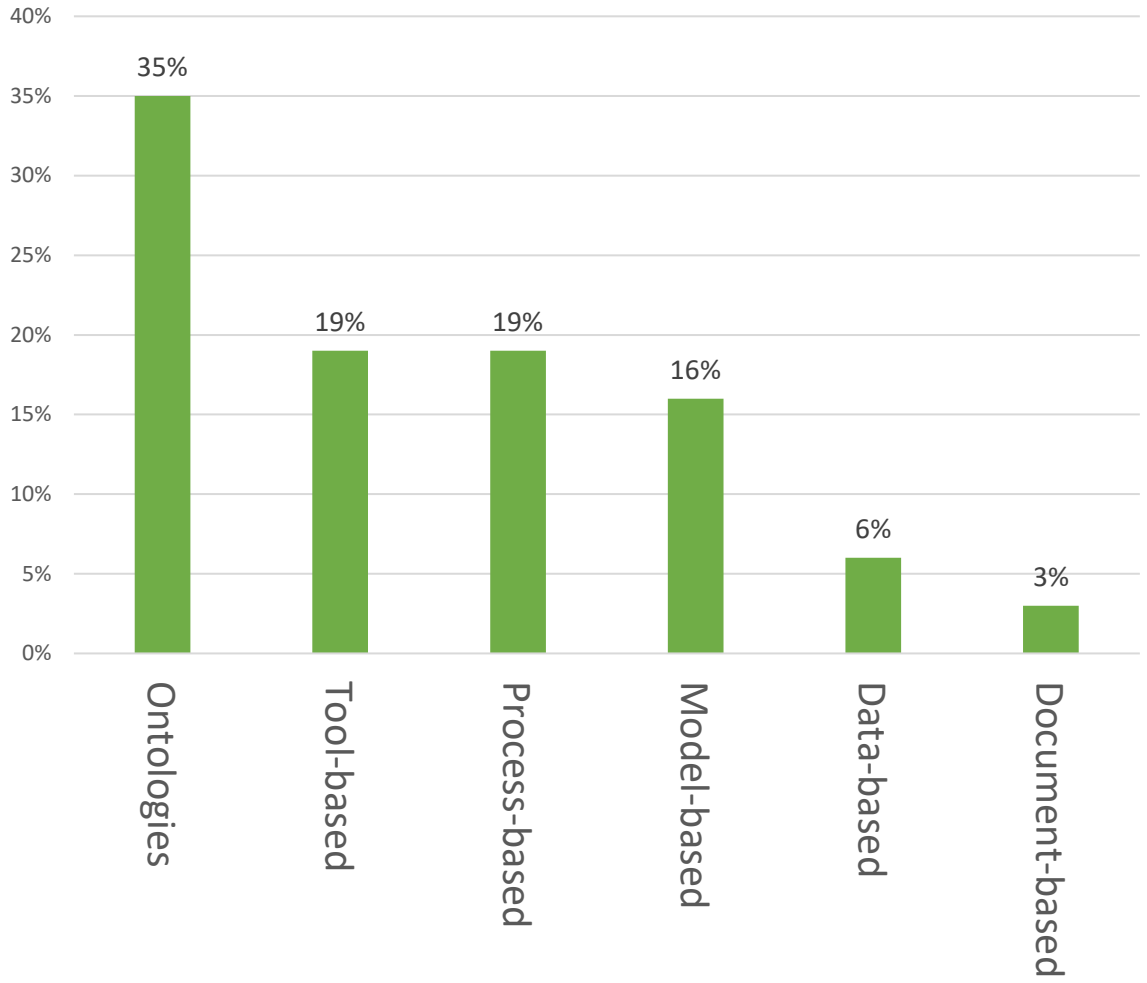
- Project (46%), Product (24%), Program (6%), Organization (24%)
- The focus on the project/product level helps them address domain-specific demands.
- Comments from participants:
  - The development of an organization-wide MBSE methodology was possible through “guidance and standards”.
  - “The aim [of having an organization-level methodology is] to facilitate reuse of digital engineering platforms across programs and projects”.

Do you find it easy to customize your MBSE tool(s) for your method?



- Tool customization is a complex task due to:
  - Lack of semantics
  - Time consuming
  - Vendor-lock solutions
  - Interoperability between tools
  - Cost is another barrier

How do you integrate different domains and disciplines in your MBSE method?



- Integration of multiple domains and disciplines emerged as a core challenge in the survey
- Only 33% of respondents provided an answer
- Using ontologies as a logical foundation or by aligning with higher-order ontologies.
- Tools mentioned were Ansys ModelCenter, Dassault Experienc, CATIA, Magic (Cameo), OpenMBEE, etc.
- Integration of multiple disciplines around a central data store with well-defined semantics

Does your MBSE method enable effective collaboration on models between your practitioners?

- 38% of respondents replied positively to the question regarding collaboration support in MBSE.

- Key Challenges:
  - Semantic and technological interoperability between tools.
  - Lack of a System-of-Systems approach in current methodologies.
  - Unclear return on investment for MBSE.
  - Difficulty in collaborative model creation and synchronizing updates across users.
  - Issues such as external stakeholder access and intellectual property concerns.
  - Wide diversity in model-specific and domain-specific practices that hinder standardization and collaboration.

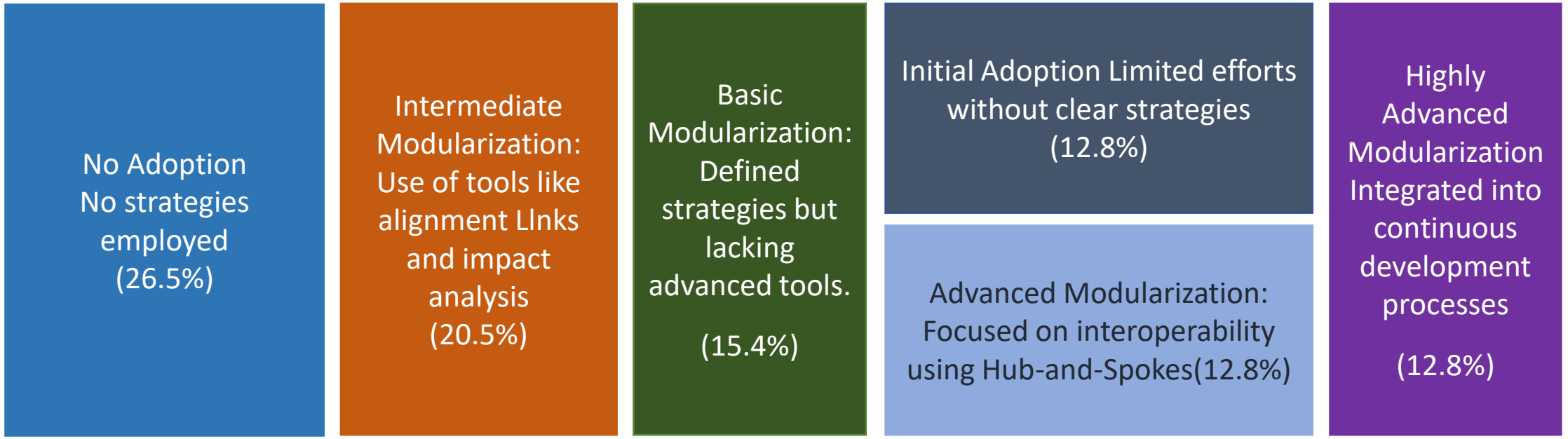
How do you give stakeholders visibility to your MBSE work? how do you review progress?

- 42% of participants reported that their MBSE methodology provided stakeholder visibility and supported reviews.
- Common practices: model-based reviews, model navigation via web pages and reports, sharing of diagrams, screenshots, etc.
- CI/CD and design science methodologies were reported, along with a focus on engaging system engineers in the stakeholder reporting.
- The MBSE experts surveyed had a strong trend towards Web-based, viewpoint-specific reporting.

Do you have a rigorous change process for your MBSE models?

- Only 14% of respondents answered positively, noting limitations due to poor tooling.
- The common practices: documented release processes, change requests, semantic versioning and modular packaging, audit logging, integrating
- CI/CD and automated reporting and using Git commit and branching strategies were mentioned.
- A trend in the expert opinions on this topic is that for rigorous change management process “the primary challenges tend to be human rather than with respect to tooling”.

Do you organize your model into separate but inter-related modules? How do you enable the co-evolution of those modules and deal with change impact?



Other Challenges

- Training and knowledge gaps (18.9%):
- Organizational resistance (18.9%)
- Technical challenges (16.2%)
- Management resistance & ROI concerns (10.8%)
- Transitioning from conventional methods (10.8%)
- Cost vs. benefit concerns (8.1%)
- Data governance and security (2.7%)

Key Recommendations

Enterprise-wide MBSE Adoption and Standardization

- Alignment of MBSE with strategic business goals
- MBSE governance structures including KPIs for performance and adoption tracking.
- Tool ecosystem development

Cross-Domain Integration

- Ontologically-aware tools for semantics
- Purpose-driven, incremental integration strategies.
- Use of standard languages (SysML v2) for enhanced semantics

Model Management, Evolution and Stakeholder Visibility

- Integrated tools with stakeholder-specific views and live reporting
- Infrastructure and collaborative processes
- Rigorous modularization and impact-analysis frameworks

Acknowledgement

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Contact the authors



Maged Elaasar
NASA JPL
maged.e.elaasar@jpl.nasa.gov



Bentley Oakes
Polytechnique Montreal
bentley.oakes@polymtl.ca



Wahab Hamou-Lhadj
Concordia University / NASA JPL
wahab.hamou-lhadj@Concordia.ca



Mohammad Hamdaqa
Polytechnique Montreal
mhamdaqa@polymtl.ca

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Thank You

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